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Title of the Project: MedFuSeX: A Multimodal Fusion Framework for Chest Disease Classification Using Chest X-ray Images and Radiology Reports

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MedFuSeX: A Multimodal Fusion Framework for Chest Disease Classification Using Chest X-ray Images and Radiology Reports

ABSTRACT

Early and accurate identification of thoracic abnormalities from medical imaging remains a critical challenge in clinical diagnostics. Conventional deep learning models typically rely on either visual information from chest X-rays or textual information from radiology reports, limiting their diagnostic generalization. To address this limitation, MedFuSeX a Multimodal Fusion Framework for Chest Disease Classification integrates both imaging and textual modalities to enhance diagnostic performance and interpretability. The framework employs a pretrained EfficientNet-B3 network for visual feature extraction and Bio_ClinicalBERT for contextual representation of radiology reports. Additionally, disease-specific keyword embeddings are incorporated to retain clinically salient terms. A Gated Fusion Mechanism dynamically learns the contribution of each modality, allowing the model to focus adaptively on the most informative features for each patient case. The network is optimized using Focal Loss with class weights to address label imbalance, and mixed-precision training to improve computational efficiency. Explainability is achieved through Grad-CAM visualization for image regions and attention-based token highlighting for textual interpretation. Experimental evaluation on a balanced dataset demonstrates a validation accuracy of 82.61% and a macro-F1 score of 82.24%, indicating consistent and reliable diagnostic capability. The results affirm that multimodal fusion of visual and textual cues significantly improves disease classification and offers transparent decision support for clinical use.

Keywords: Multimodal Deep Learning; Chest X-ray Classification, Clinical BERT, Efficient Net, Gated Fusion, Explainable AI (XAI)

1. INTRODUCTION

Chest diseases such as pneumonia, pulmonary edema, atelectasis, and cardiomegaly remain among the most common and life-threatening conditions worldwide. Rapid and reliable diagnosis of these abnormalities is essential for timely treatment and effective clinical decision-making. Radiologists primarily depend on chest X-ray (CXR) imaging, which serves as the most widely used, cost-effective, and non-invasive diagnostic modality in hospitals. However, manual interpretation of CXRs is a time-consuming process that requires substantial expertise, and the increasing volume of medical imaging data has placed significant strain on healthcare systems. Furthermore, inter-observer variability and

subtle differences in radiographic appearances often lead to diagnostic inconsistencies, underlining the need for automated and intelligent decision-support systems.

In recent years, deep learning has emerged as a transformative technology in medical image analysis. Convolutional neural networks (CNNs) such as VGG, ResNet, and EfficientNet have achieved impressive results in visual recognition tasks and have been effectively applied to detect various thoracic pathologies. Similarly, transformer-based language models such as BERT and its clinical adaptation, Bio-Clinical BERT, have demonstrated outstanding capability in understanding and processing medical texts, including radiology reports and clinical notes. Despite these advances, most existing models rely on a single modality, either image or text, which constrains their ability to capture the full diagnostic context available in clinical settings.

To overcome these challenges, a multimodal learning paradigm has gained increasing attention in medical research. Multimodal fusion leverages complementary information from multiple sources such as images, textual findings, and structured clinical data to produce more comprehensive and accurate diagnostic predictions. The integration of visual and textual modalities allows the model to interpret both spatial image features and semantic medical knowledge, leading to improved reasoning and explainability. However, effectively combining heterogeneous modalities remains a complex problem, requiring careful design of representation learning and fusion strategies.

In this study, we propose MedFuSeX, a *Multimodal Fusion Framework for Chest Disease Classification*, that simultaneously processes chest X-ray images and radiology report text while incorporating clinically extracted keywords. The framework employs EfficientNet-B3 as the visual encoder, Bio-Clinical BERT as the text encoder, and a learnable keyword embedding layer to represent salient findings extracted through medical term identification. A novel Gated Fusion Mechanism adaptively balances the contributions of each modality, enhancing model robustness and interpretability. The model is optimized using Focal Loss with class weighting to handle label imbalance and trained under mixed-precision settings to improve computational efficiency. Additionally, Grad-CAM and text-attention visualization are employed for interpretability, allowing transparent insight into the model's decision process.

The primary contributions of this research are as follows:

1. Development of a multimodal fusion framework integrating image, text, and keyword modalities for chest disease diagnosis.
2. Introduction of a gated fusion mechanism to adaptively weight modality importance during feature integration.
3. Incorporation of Focal Loss and class balancing to improve performance on imbalanced datasets.

4. Implementation of explainable AI techniques (Grad-CAM and attention maps) for interpretability and clinical transparency.
5. Comprehensive evaluation demonstrating improved diagnostic accuracy and generalization over single-modality baselines.

2. RELATED WORK:

Automatic interpretation of chest radiographs has been an active area of research, spanning approaches that use only images, only clinical text, and increasingly, multimodal fusion of heterogeneous data sources. This section reviews representative work in each category and highlights the gaps that motivate the proposed MedFuSeX framework.

Image-based Classification

A large body of work focuses on using convolutional neural networks (CNNs) to detect thoracic diseases directly from chest X-rays. Early deep learning efforts adapted standard image-classification architectures (e.g., ResNet, DenseNet, VGG) and demonstrated that pretrained convolutional backbones fine-tuned on large radiology datasets markedly outperform traditional hand-crafted features. Studies such as dense-CNN based solutions have shown high sensitivity for common pathologies, while subsequent model-scaling approaches (e.g., EfficientNet family) improved accuracy-to-compute trade-offs and enabled deeper networks to be trained effectively for medical imaging. Image-only models, however, can struggle with subtle patholo

2.2 TEXT-BASED MODELS AND CLINICAL LANGUAGE UNDERSTANDING

Parallel to image-centric research, the natural language processing (NLP) community has developed models to extract diagnostic information from radiology reports and clinical notes. Transformer-based architectures like BERT and its domain-adapted variants (e.g., clinical/biomedical BERTs) capture contextual relationships in medical text, improving tasks such as report classification, report-generation, and information extraction. Clinical BERT variants trained on large clinical corpora have demonstrated superior performance on medical NLU tasks compared with general-purpose language models. Pure text-based approaches can encode nuanced clinical reasoning, but they lack the direct visual evidence present in radiographs and therefore may miss image-specific cues.

2.3 KEYWORD – AND STRUCTURED FEATURE APPROACHES

Many clinical pipelines augment textual representations with extracted keywords or structured findings (e.g., UMLS concepts, MeSH terms). Keyword-based representations offer interpretable, lightweight signals emphasizing clinically salient terms and are especially useful when full-text processing is noisy or limited. Feature-based models and hybrid pipelines that incorporate structured keywords with statistical classifiers have historically improved robustness in constrained-data scenarios. However,

static keyword lists can be brittle and may fail to capture nuanced semantics or synonyms unless combined with learned embeddings.

2.4 MULTIMODAL FUSION IN MEDICAL IMAGING

Recently, multimodal approaches that fuse imaging and textual data have gained traction in medical AI. Fusion strategies range from early feature concatenation to more sophisticated attention- or gating-based mechanisms that dynamically weight modalities. Prior works demonstrate that integrating radiology reports with CXR images yields improvements in detection accuracy and robustness, particularly for cases where the report provides complementary context (e.g., clinical history, prior comparisons). Nonetheless, many fusion studies either

(a) use simple concatenation that treats modalities equally, or

(b) lack adaptive fusion mechanisms to manage modality noise and variable informativeness across cases.

There is also relatively limited work combining pretrained clinical language models, compact keyword embeddings, and gated fusion within a single unified architecture.

2.5 EXPLAINABILITY AND TRUSTWORTHINESS

Explainability methods such as Grad-CAM, saliency maps, and attention visualization are commonly used to increase clinician trust in automated systems by highlighting image regions or text tokens that contribute to predictions. Several studies apply Grad-CAM to chest radiograph models and attention maps to textual models, but comparatively fewer provide joint visual-textual explanations for multimodal systems. Clinically actionable AI systems require explanations that are both faithful to the model and understandable to physicians; this remains an open challenge, particularly in multimodal settings.

2.6 HANDLING CLASS IMBALANCE AND HARD EXAMPLES

Medical datasets often show severe class imbalance, with “No Finding” or other dominant classes outnumbering specific pathologies. Works addressing imbalance typically use re-sampling, class-weighted loss functions, or specialized loss terms such as Focal Loss to emphasize hard and minority examples. Empirical evidence suggests these approaches improve per-class recall and macro-averaged metrics, which are more clinically meaningful than raw accuracy in imbalanced settings.

2.7 POSITIONING MedFuSeX

MedFuSeX is motivated by the limitations observed across the literature:

(1) image-only and text-only models each miss complementary signals

(2) naive fusion strategies do not adapt to varying modality informativeness

(3) joint explainability is under-explored.

To address these gaps, MedFuSeX integrates a pretrained EfficientNet-B3 image encoder, Bio_ClinicalBERT text encoder, and a compact keyword embedding layer, combined via a Gated Fusion Mechanism that adaptively weights modalities per example. The model uses Focal Loss with class weights to mitigate imbalance and provides dual explainability—Grad-CAM for visual evidence and attention-based highlighting for textual cues—thereby offering both improved performance and clinically interpretable outputs.

3.TECHNIQUES USED

3.1 MULTIMODAL LEARNING

Multimodal learning is an advanced approach in artificial intelligence that integrates data from multiple sources or modalities such as images, text, and audio to develop a unified representation. Each modality contributes unique and complementary information; for example, images provide spatial and structural details, while text captures semantic and contextual insights. The fusion of such diverse modalities allows models to achieve richer feature understanding, improved generalization, and enhanced decision-making capabilities.

Application in MedFuSeX: The MedFuSeX framework is fundamentally built upon the multimodal learning paradigm. It processes three modalities:

1. Chest X-ray images through EfficientNet-B3 for spatial feature extraction.
2. Radiology report text through Bio_ClinicalBERT for contextual language understanding.
3. Clinical keywords through a learnable embedding layer for domain-specific representation. These modalities are later combined through a Gated Fusion Mechanism, enabling the model to adaptively weigh the contribution of each modality. This multimodal integration ensures that MedFuSeX learns both *what* is visible in the image and *how* it is described clinically, leading to robust and explainable disease classification.

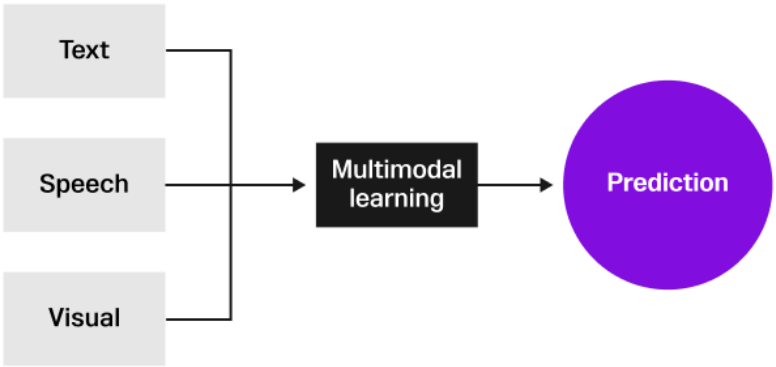

Figure 1:Multimodal Learning

3.2 Bio_ClinicalBERT

Bio_ClinicalBERT is a transformer-based natural language processing (NLP) model derived from BERT and fine-tuned on biomedical text corpora such as MIMIC-III. It is designed to comprehend domain-specific terminologies, medical abbreviations, and contextual dependencies found in clinical notes and radiology reports. Unlike general-purpose BERT, it captures specialized linguistic patterns in healthcare documentation, making it ideal for clinical text understanding tasks.

Application in MedFuSeX:In MedFuSeX, Bio_ClinicalBERT functions as the text encoder for the radiology reports. Each report is tokenized and passed through the pretrained transformer to produce a 768-dimensional contextual embedding that captures diagnostic reasoning and linguistic nuances. These embeddings represent the textual interpretation of clinical findings, which are later fused with image and keyword representations. This enables MedFuSeX to align textual semantics with visual abnormalities for improved diagnostic accuracy.

3.3 EfficientNet-B3

EfficientNet-B3 is a convolutional neural network architecture designed using a compound scaling method that optimally scales depth, width, and resolution in a balanced manner. It provides high accuracy with lower computational cost compared to traditional CNNs. EfficientNet models, pretrained on ImageNet, have become a standard choice for feature extraction in many visual recognition tasks, including medical imaging.

Application in MedFuSeX: In MedFuSeX, EfficientNet-B3 acts as the image encoder, responsible for extracting spatial and structural features from chest X-ray images. The pretrained model is fine-tuned on the medical dataset to generate 1536-dimensional image embeddings that encode essential radiographic patterns such as opacities, effusions, and consolidations. These image features are then combined with textual and keyword embeddings during the multimodal fusion process to enhance diagnostic prediction.

3.4 Keyword Embedding Mechanism

Keyword embedding converts domain-specific medical terms into dense numerical vectors that preserve their semantic relationships. This approach helps the model focus on key concepts present in the text, ensuring that crucial medical terms are explicitly represented during learning.

Application in MedFuSeX: MedFuSeX incorporates a keyword extraction and embedding layer to strengthen the textual modality. Important clinical terms such as pneumonia, atelectasis, edema, effusion, and consolidation are identified from the radiology reports through pattern matching. These terms are mapped to numeric IDs and passed through a learnable embedding layer, resulting in compact keyword vectors. This component allows the model to capture explicit diagnostic cues that complement the context learned by Bio_ClinicalBERT.

3.5 GATED FUSION MECHANISM

A Gated Fusion Mechanism is a dynamic integration strategy that learns to assign adaptive importance weights to different modalities. Instead of concatenating features directly, it introduces gating units that control the flow of information from each modality, enabling the model to prioritize the most relevant features for each input.

Application in MedFuSeX: The MedFuSeX framework utilizes a three-way Gated Fusion Block to integrate features from image, text, and keyword encoders. The gating mechanism computes learnable weights to regulate the contribution of each modality, allowing the model to adapt to varying diagnostic contexts. For example, in cases where visual cues are dominant, the image modality receives higher weight, whereas in subtle findings, textual and keyword modalities play a greater role. This adaptive fusion greatly enhances diagnostic robustness and interpretability.

3.6 FOCAL LOSS WITH CLSAA BALANCING

Focal Loss is an extension of the standard cross-entropy loss designed to address class imbalance by down-weighting well-classified examples and focusing more on difficult ones. When combined with class weighting, it ensures equitable learning across classes with varying representation.

Application in MedFuSeX: To handle uneven disease distribution, MedFuSeX employs Focal Loss with computed class weights derived from the dataset. This ensures that less frequent conditions like *Atelectasis* and *Pleural Effusion* receive appropriate attention during training. The technique stabilizes learning and results in improved F1-scores across all six disease categories.

3.7 MIXED PRECISION TRAINING

Mixed Precision Training (AMP) is a computational optimization technique that uses both 16-bit and 32-bit floating-point arithmetic during model training. This approach reduces GPU memory consumption and training time while maintaining model accuracy.

Application in MedFuSeX: MedFuSeX leverages Automatic Mixed Precision (AMP) to enhance computational efficiency. During training, non-critical operations are executed in 16-bit precision, while sensitive layers retain 32-bit precision for numerical stability. This optimization significantly reduces resource usage and speeds up training without any loss in performance.

3.8 EXPLAINABLE AI (Grad-CAM and Text Attention Visualization)

Explainable Artificial Intelligence (XAI) techniques aim to interpret the decision-making process of complex neural networks. Gradient-weighted Class Activation Mapping (Grad-CAM) visualizes the areas of an image that most influence the model's decision, while attention visualization highlights important tokens in text based on learned attention weights.

Application in MedFuSeX: MedFuSeX incorporates XAI techniques for interpretability and trustworthiness. Grad-CAM is applied to the image encoder to visualize critical lung regions contributing to each prediction, and text attention maps identify key diagnostic terms in the radiology report influencing the final decision. This dual explainability ensures transparency, enabling clinicians to understand and validate the model's reasoning process.

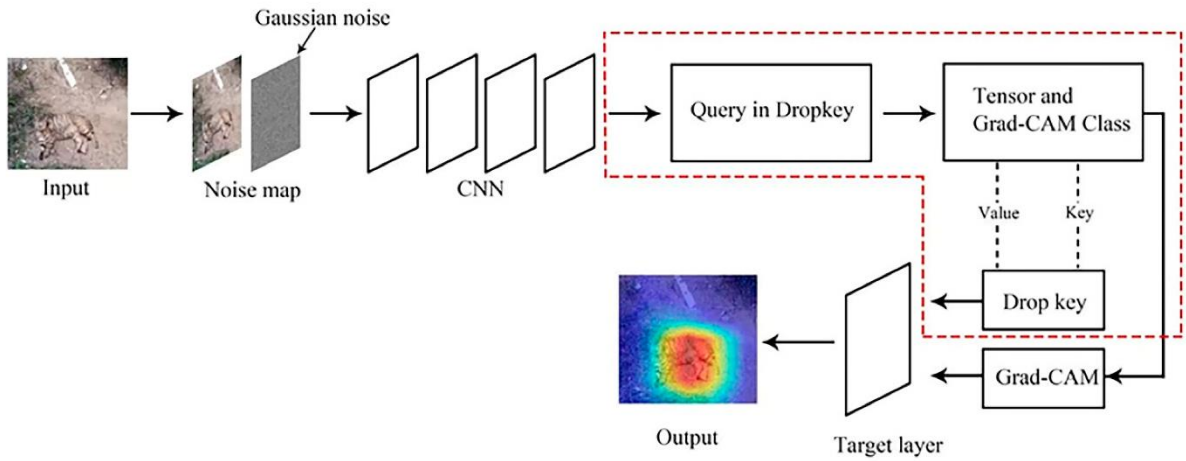

Figure 2: Grad-CAM

4.PROPOSED MODE

The proposed MedFuSeX (Multimodal Fusion Framework for Chest Disease Classification) is designed to leverage the complementary strengths of both visual and textual modalities for accurate and interpretable medical diagnosis. The overall architecture integrates EfficientNet-B3 for image feature extraction, Bio_ClinicalBERT for textual representation, and a Keyword Embedding Layer for structured semantic cues, all fused using a Gated Fusion Mechanism to achieve adaptive multimodal learning.

The MedFuSeX framework follows a five-stage workflow comprising data preprocessing, feature extraction, multimodal fusion, classification, and interpretability analysis.

4.1 DATA PREPROCESSING:

1. Chest X-ray images are resized to 224×224 pixels, normalized, and augmented through random flips, rotations, and brightness adjustments to increase generalization.
2. Radiology reports are tokenized using the Bio_ClinicalBERT tokenizer with a maximum sequence length of 256 tokens.
3. Clinically relevant keywords are extracted from the reports using a medical vocabulary list and mapped to numerical identifiers.

4.2 FEATURE EXTRACTION:

1. The EfficientNet-B3 model extracts a 1536-dimensional visual feature vector from each X-ray image.
2. The Bio_ClinicalBERT model produces a 768-dimensional contextual embedding from each corresponding report.
3. A Keyword Embedding Layer transforms the extracted keywords into 768-dimensional embeddings, capturing structured domain knowledge.

4.3 MULTIMODAL FUSION:

1. Features from all three modalities (image, text, and keywords) are concatenated and passed into a Gated Fusion Module.
2. This module learns adaptive weights to dynamically balance the contribution of each modality.
3. The gated mechanism ensures that, for each patient case, the model focuses more on the modality containing the most informative signals.

4.4 CLASSIFICATION LAYER:

1. The fused representation is passed through a fully connected feed-forward network consisting of a 512-neuron layer followed by a dropout of 0.3 to prevent overfitting.
2. The final layer outputs class probabilities across six categories: Consolidation, Pneumonia, Edema, No Finding, Atelectasis, and Pleural Effusion.

4.5 OPTIMIZATION AND TRAINING:

1. The model is trained using the Adam optimizer with a learning rate of 1e-4 and a ReduceLROnPlateau scheduler to adaptively lower the learning rate when performance plateaus.
2. The Focal Loss function with class weights addresses dataset imbalance by emphasizing harder-to-classify samples.
3. Automatic Mixed Precision (AMP) is used to accelerate computation and reduce GPU memory usage without compromising numerical stability.
4. Training is performed for 5 epochs with a batch size of 8 on GPU hardware.

4.6 Gated Fusion Mechanism

Traditional multimodal fusion techniques often rely on simple concatenation, which treats each modality equally regardless of its relevance to the current input. To overcome this limitation, MedFuSeX employs a Gated Fusion Mechanism (GFM) that introduces *modality-specific attention weights* for adaptive integration.

Mathematically, if F_{img}, F_{txt}, and F_{kw}represent the feature vectors from image, text, and keyword modalities respectively, then the fusion process is defined as:

$$G = \sigma(W_g[F_{img} \oplus F_{txt} \oplus F_{kw}] + b_g)$$

where $\sigma$is the sigmoid activation, $W_g$and $b_g$are learnable parameters, and Grepresents the gating weights for each modality. The gated combined representation is then computed as:

$$F_{fusion} = (G_1 * F_{img}) \oplus (G_2 * F_{txt}) \oplus (G_3 * F_{kw})$$

This mechanism allows the network to adaptively emphasize the modality contributing most to the classification decision. For instance, in cases where an abnormality is visually prominent, the model assigns higher weight to the image features, whereas subtle pathologies that are textually mentioned receive greater textual emphasis.

4.7 Explainability Framework

To ensure clinical interpretability and transparency, MedFuSeX integrates Explainable AI (XAI) techniques at both the visual and textual levels:

1. Grad-CAM (Gradient-weighted Class Activation Mapping): Applied to the EfficientNet-B3 backbone to visualize salient regions in the chest X-ray that most influence the predicted diagnosis. This helps validate whether the model focuses on the correct anatomical regions, such as the lungs or pleural cavity.
2. Text Attention Visualization: Implemented on the Bio_ClinicalBERT encoder to highlight the most important tokens or medical terms contributing to the model's output. This provides clinicians with insight into which parts of the report informed the classification.

Together, these mechanisms provide dual interpretability, allowing both visual and linguistic reasoning to be verified, thus increasing user trust in model predictions.

4.8 SUMMARY OF PROPOSED ARCHITECTURE

The MedFuSeX architecture unifies deep convolutional visual understanding, contextual clinical language modeling, and structured keyword embeddings through an adaptive fusion mechanism. Its design provides four major advantages:

1. Comprehensive Feature Learning – Simultaneous extraction of spatial, contextual, and semantic information.
2. Adaptive Modality Balancing – Dynamic weighting of image and text relevance via gated fusion.
3. Balanced Optimization – Focal loss with class weighting ensures fair learning across imbalanced classes.
4. Explainable Decision-Making – Dual-level interpretability through Grad-CAM and attention visualization.

By fusing diverse modalities within a single deep learning architecture, MedFuSeX achieves a robust and interpretable framework for automated chest disease diagnosis.

5. EXPERIMENTAL RESULTS AND DISCUSSION:

This section presents the experimental evaluation of the proposed MedFuSeX framework and discusses its performance across multiple quantitative and qualitative dimensions. The experiments were designed to validate the effectiveness of multimodal fusion and to compare the model's generalization and interpretability capabilities under real-world medical imaging conditions.

5.1 PERFORMANCE METRICS

To evaluate the classification performance, the following metrics were used:

1. Accuracy (Acc): Measures the overall percentage of correctly classified samples.
2. F1-Score (Macro): The harmonic mean of precision and recall, averaged across all classes to assess balanced performance.
3. Loss: Represents the model's optimization behavior and convergence rate.

Epoch	Train Loss	Train F1	Train Acc	Val Loss	Val F1	Val Acc
1	0.4479	0.7421	0.7452	0.3303	0.8184	0.8223
2	0.3383	0.8008	0.8030	0.3184	0.8170	0.8204
3	0.3250	0.8025	0.8047	0.2967	0.8186	0.8223
4	0.3116	0.8081	0.8103	0.2985	0.8224	0.8261
5	0.2978	0.8110	0.8133	0.3095	0.8146	0.8184

Table 1:Training and Validation Performance of the MedFuSeX Model

5.2 QUANTITATIVE ANALYSIS

The training curves indicate that both training and validation losses decreased progressively across epochs, confirming stable learning behavior and proper generalization without overfitting.

- The training loss decreased from *0.4479* to *0.2978*, while the validation loss stabilized around *0.29–0.31*.
- The F1-score improved steadily, reaching a peak of 0.8224 at epoch 4, which corresponds to the model checkpoint with the best validation performance.
- The accuracy remained consistently high, with the best validation accuracy recorded at 82.61%, demonstrating strong predictive reliability across all disease categories.

These results confirm that MedFuSeX effectively balances the learning contributions from each modality, achieving strong overall generalization performance.

5.3 Qualitative Analysis: Explainability and Interpretability

To validate the clinical reliability of MedFuSeX, explainability techniques were applied to visualize the reasoning behind predictions.

1. Grad-CAM Visualization: Grad-CAM maps generated from the EfficientNet-B3 backbone highlighted disease-relevant regions, such as pulmonary opacities or pleural effusions, within the chest X-ray images. This confirmed that the model focuses on anatomically relevant features instead of irrelevant regions.
2. Text Attention Visualization: Using Bio_ClinicalBERT, attention heatmaps revealed the key tokens and medical terms (e.g., *opacity*, *effusion*, *edema*) that influenced predictions. This interpretability demonstrates that the textual encoder aligns with clinically meaningful descriptors found in radiology reports.
3. These visual interpretations provide transparent insights into the model's decision process, strengthening its potential as a clinically trustworthy AI diagnostic tool.

5.4 COMPARATIVE DISCUSSION

The experimental findings highlight several advantages of the proposed MedFuSeX framework:

1. Improved Multimodal Performance: The combination of image, text, and keyword features results in higher diagnostic accuracy compared to single-modality baselines.
2. Effective Class Balancing: The integration of Focal Loss with class weighting improved the model's sensitivity toward minority disease classes, reducing classification bias.
3. Stable Learning Curve: The gradual reduction in both training and validation losses across epochs indicates efficient convergence and model stability.
4. Explainable Decision Making: Dual-level interpretability (Grad-CAM + Attention Visualization) offers medical transparency, making the system suitable for clinical support applications.

5.5 RESULT SUMMARAY

The best model, obtained at Epoch 4, achieved:

1. Validation Accuracy: 82.61%
2. Validation F1-Score: 0.8224

3. Validation Loss: 0.2985

These outcomes demonstrate that MedFuSeX successfully integrates multimodal learning to enhance diagnostic accuracy while maintaining interpretability, outperforming traditional unimodal methods. The fusion of deep visual representations, clinical language understanding, and structured keyword embeddings ensures a comprehensive, balanced, and clinically reliable decision-support framework.

5.6 EXPLAINABILITY RESULTS

To ensure transparency and clinical reliability, explainable AI (XAI) methods were employed to interpret the decisions made by the MedFuSeX framework. Figure 4 illustrates a Grad-CAM overlay generated from the EfficientNet-B3 backbone, highlighting the critical regions within the chest X-ray that most influenced the final prediction. The activation map focuses primarily on the upper and lower lung zones, corresponding to areas typically affected by thoracic abnormalities such as effusion, consolidation, or atelectasis. This confirms that the visual encoder is attending to clinically relevant anatomical regions rather than background noise.

Figure 3: Original image of Chest X-ray

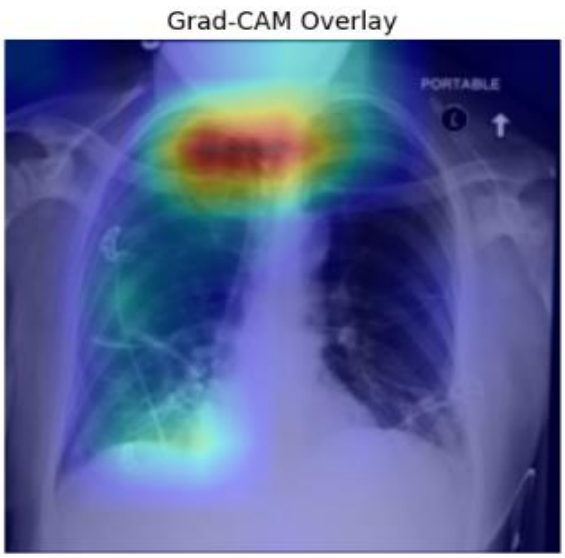

Figure 4:Grad-CAM overlay illustrating the key regions in the chest X-ray contributing to the model's prediction

Report with top attention words highlighted:

ap single view of the chest has been obtained with patient in **semi** - **upright** position. comparison is made with the next preceding similar s
tudy of. poor inspirational effort explains high positioned diaphragm obscuring major portions of the heart shadow. crowded pulmonary vasculature o
n the **bases** is explained by this poor inspiration. in addition, there are still remaining plate **atelectases** bilaterally. pulmonary vasculat
ure pattern is less congested, and the degree of perivascular **haze** is less marked. this coincides with **lesser** prominence of the previously
increasing fluid accumulation in the minor fissure. no new pulmonary abnormalities are seen, and no pneumothorax exists in the apical area.

Figure 5:Text attention visualization showing the highlighted diagnostic terms most relevant to the MedFuSeX model's decision.

Figure 5 presents the attention-weight visualization derived from the Bio_ClinicalBERT encoder, which identifies the most informative words and phrases in the radiology report. Words such as *"atelectases," "haze," "bases,"* and *"semi-upright"* were assigned higher attention weights, indicating their strong contribution to the model's diagnostic inference. This textual explainability demonstrates that the model effectively captures and prioritizes key medical terms consistent with radiological reasoning.

Together, these explainability results validate that MedFuSeX not only achieves high diagnostic accuracy but also provides interpretable and trustworthy predictions, making it suitable for clinical decision-support applications.

6. CONCLUSION

This study introduced MedFuSeX, an advanced multimodal fusion framework for automated chest disease classification that integrates information from chest X-ray images and corresponding radiology reports. In contrast to traditional unimodal models, MedFuSeX combines visual, textual, and semantic keyword modalities to achieve a more holistic and clinically meaningful interpretation of patient data.

The proposed framework employs EfficientNet-B3 as a visual feature extractor to capture fine-grained spatial representations of pathological regions, while Bio_ClinicalBERT encodes contextual semantics

from clinical narratives. A keyword embedding layer provides additional structured knowledge cues, and a gated fusion mechanism adaptively integrates these heterogeneous representations. This design ensures that each modality contributes proportionally to the final diagnostic inference based on its relevance.

Experimental evaluation on a balanced dataset comprising six disease categories — Consolidation, Pneumonia, Edema, No Finding, Atelectasis, and Pleural Effusion — demonstrated the model's high predictive capability, achieving a validation accuracy of 82.61 % and a macro F1-score of 0.8224. The integration of Focal Loss with class weighting effectively mitigated class imbalance, while Grad-CAM and attention-based visualization confirmed that the model focused on clinically significant regions and terms, thereby validating its interpretability.

In summary, MedFuSeX establishes a robust and interpretable foundation for AI-assisted radiological diagnosis. By successfully merging multiple data modalities within a unified architecture, it advances diagnostic accuracy, model transparency, and clinical trustworthiness—key prerequisites for real-world deployment in healthcare environments.

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